

Data Mining and Machine Learning Course Outline

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Course Overview

- Master's-level course with 12 weeks of instruction, 6h/week in 3x2h blocks.
- Designed as a toolbox of data analysis methods.
- Programming in Python using VS Code.
- 20% mid-term, best grade out of 2.
- 20% project reports. 4 people per group max, so minimum circa 25 reports.
- 20% computer exercises. Rolling hand-over (W1 exercises has to be handed in at the end of W2).
- 40% final exam.

Weekly Breakdown

Week 1: Setup and Data Preparation

Lecture 1: Environment setup (Python, VS Code).

- Course structure.
- Installing requirements in Python environments.

Lecture 2: Data loading, cleaning, and EDA.

- Handling missing values, encoding, normalization.
- Descriptive statistics and visualization.

Computer Exercise:

- Titanic: clean, describe, visualize.
- Practice exploratory analysis and preprocessing.

Week 2: Time Series with ARIMA

Lecture 1: Time series fundamentals.

- Trend, seasonality, stationarity.
- Autocorrelation, differencing.

Lecture 2: ARIMA modeling.

- AR, MA, and integration components.
- Forecasting and residual analysis.

Computer Exercise:

- Airline passengers dataset: deal with datatype.
- Build and evaluate ARIMA: plot forecasts, analyze residuals.

Week 3: Regression and Classification Basics

Lecture 1: Regression models.

- Linear (and polynomial regression).
- Bias-variance tradeoff (visual).

Lecture 2: Classification models.

- Logistic regression and k-NN.
- Evaluation metrics: confusion matrix, precision, recall, F1.

Computer Exercise:

- Housing dataset for regression, Iris for classification.
- Train models, evaluate performance, interpret outputs.

Week 4: SVMs and Model Tuning

Lecture 1: Support Vector Machines.

- Max-margin intuition, kernel trick.
- RBF kernel, non-linear classification.

Lecture 2: Cross-validation and hyperparameter tuning.

- Grid search, model selection, data leakage.
- Pipeline design.

Computer Exercise:

- Digits dataset: train SVM with linear and RBF kernel.
- Perform cross-validation and grid search for best parameters.

Week 5: Trees and Ensembles

Lecture 1: Decision Trees.

- Splitting criteria, overfitting, pruning.
- Interpretability and visualization.

Lecture 2: Random Forests.

- Bagging, ensemble learning.
- Feature importance and model robustness.

Computer Exercise:

- UCI dataset: build decision tree and random forest.
- Compare performance, visualize tree, analyze feature importance.

Week 6: Boosting and Advanced Tree Models

Lecture 1: Boosting concepts.

- Gradient Boosting Machines, AdaBoost vs Gradient Boosting.
- Bias reduction and sequential fitting.

Lecture 2: LightGBM, XGBoost, and CatBoost.

- Use cases, parameter tuning, efficiency.
- Interpretability and SHAP values.

Computer Exercise:

- Structured dataset (e.g., bank or housing data).
- Compare random forest vs XGBoost vs LightGBM.
- Use SHAP to interpret predictions.

Week 7: Clustering and Dimensionality Reduction

Lecture 1: Clustering.

- K-means, elbow method, limitations.
- DBSCAN, hierarchical clustering overview.

Lecture 2: PCA and feature engineering.

- Dimensionality reduction, variance explained.
- Feature selection vs feature extraction.

Computer Exercise:

- Iris dataset clustering with PCA visualization.
- Try different values of k, analyze silhouette scores.

Week 8: Midterm 1 and Neural Networks Fundamentals

Midterm 1.

- 2h, pen and paper, 1 sheet of paper (recto-verso) allowed.
- Focused on statistical methods seen up to this point. Nothing about neural networks.

Lecture 2: Neural Networks Fundamentals.

- Network architecture and forward pass.
- Gradient descent, overfitting.

Computer Exercise:

- Digits: build MLP for digit MNIST classifier.
- Experiment with dropout and optimizers.

Week 9: Training techniques and CNNs

Lecture 1: Training techniques, transfer learning and regularization.

- Dropout, BatchNorm, Adam optimizer.
- Early stopping and learning rate schedules.
- Pre-trained models, fine-tuning.

Lecture 2: CNN fundamentals.

- Convolutions, pooling, hierarchical features.
- CNN architecture and applications.

Computer Exercise:

- Fashion-MNIST: build CNN, evaluate.
- Use pre-trained model for classification.

Week 10: Sequences and Transformers

Lecture 1: RNNs and LSTMs.

- Sequence modeling, hidden state, memory.
- Applications to text and time series.

Lecture 2: Transformers and attention.

- Attention mechanism and self-attention.

- Transformer architecture, use cases.

Computer Exercise:

- Sentiment analysis with LSTM and HuggingFace Transformers.
- Compare performance and training behavior.

Week 11: Midterm 2 and Project Kickoff

Lecture 1: Midterm 2.

- 2h, pen and paper, 1 sheet of paper (recto-verso) allowed.
- Only on NN/CNN/RNN/CNN.

Lecture 2: Project kickoff.

- Expectations, datasets, problem scoping.
- Guidance on methodology and milestones.

Help with projects:

- Explore project dataset or begin case study implementation.
- Apply multiple models and evaluate.

Week 12: Case Studies 1 and 2

Lecture 1: Case Study 1 – Structured Data.

- Real-world numeric and categorical dataset.
- Complete pipeline: cleaning, feature selection, modeling.
- Weather data from Icelandic Meteorological office?

Lecture 2: Case Study 2 – Images + Wrap-up.

- CNN or transfer learning on small image set.
- Discuss trade-offs in model choice.
- Mechanical arm recognizing pieces in an assembly chain?
- Toolbox summary and flowchart for model selection. One/Two pager that resumes each use case seen in the class.
- Further learning resources and closing remarks.

Help with projects:

- Implement image or text-based model.
- Interpret predictions and performance.

Final Exam:

- 3h, pen and paper, 2 sheets of paper recto-verso.
- Covers most of the course. 2 sections: trad machine learning and NN+variations, think Midterm 1+2.